

Companion.energy case: contract management

Introduction

The case asked me to explore contract management in Companion.energy, pick a problem worth solving, and prototype a solution. I went through the product as a first-time user, then focused on one main problem: how a customer's assets and their contracts relate. A second, smaller proposition redesigns the Futures chart on Market Data.

How I worked. The thinking and the decisions are my own. I used Claude as a building partner to write the prototype's code, and I reviewed each part. The assignment took me between 4 and 5 hours. I sketched the ideas on the go in a notebook; there was no time for Figma.

First time through the dashboard

I came to the product without knowing it, the energy field, or what its customers need.

Overall, the experience felt like a maze. It was hard to know where I was, and to understand the connections between elements: the assets I created in Control Room, the contracts listed in Contracts, and how one relates to the other. Creating assets and contracts was complicated; the forms assume knowledge I did not have, and nothing confirmed that what I had built made sense.

The freemium plan added to that. It did not let me add more assets, so I could not see how the product behaves with several of them, and I was left with questions: how several sites or grid connections are shown together, how contracts spread across them, and how the diagram holds up as a setup grows.

Control Room remodelling

The problem

- **Who:** the person managing a customer's energy, especially someone new to the platform or to the field, and customers with several sites.
- **Impact:** they cannot tell which sites are covered by a contract. A site without one has costs nobody modelled, and nothing points it out. Assets live in Control Room, contracts in Contracts, and the asset view is read only.

- **Why it matters:** contracts are the bridge between physical energy flows and financial outcomes. The brief itself notes that nothing is validated against reality; a missing contract is the plainest case.

Directions considered

- **A list or tree of sites with their coverage:** simple, and fits a single-site customer, but it does not show where sites are and turns flat once a portfolio grows.
- **An editable asset view on the Contracts page:** it fixes the read-only view, but keeps two pages, so an asset and its contracts stay apart.
- **Merging Control Room and Contracts around a map (chosen):** one place answers "what is this site, and what pays for it?"

Why the map

The idea came from management games like SimCity, where you run a whole energy system from one screen. A map identifies and locates every site, scales from one site to many through clusters, and shows at a glance the sites without a contract. The trade-off: a single-site customer gains little from a map, but the same site panel works from a plain list.

What the prototype shows

- **Map:** 26 sites in Belgium. Each marker shows what the site is (industrial plant, warehouse, solar farm...) and a badge for one or several contracts; no badge means no contract. Nearby sites group into a counter that zooms in on press.
- **Site diagram:** selecting a site opens its grid connection and the assets behind it. The grid connection card gives the number of contracts. The map shrinks to an embed in the corner, and the selected site stays visible, in the primary colour, even inside a cluster.
- **Batteries:** the icon shows the charge (red, orange, green) and animates while charging. The card says what the battery does in plain words: Charging, Supplying, Standby, rather than Discharging and Idle.
- **Contracts page:** disabled for now; contracts are reached through their sites.

Going further

- **Open the contracts from the grid connection:** name, supplier, validity and each line in plain language, with gaps flagged, for example solar injection with no selling line.
- **Live flows on the links:** animated arrows showing which way energy moves, the power in kW, and the price next to each battery's state, so its cost or earnings read on its own card.

- **Alerts on the map:** an outage, a price too high or not the one the contracts lead to expect, a budget exceeded. Maintenance by local companies could feed the physical state: the next check-up and the result of the last one.
- **Assets in detail, not in bulk:** a contract today sees a site's solar panels as one asset. Seeing each installation shows how many there are, what each produces, and which ones a contract covers; a PPA is tied to a specific producing asset.

Market Data: Futures

The problem

The chart draws 18 lines (6 delivery months × base, peak and off-peak) in a narrow price range. Colour means neither month nor profile, the legend is paginated, the tooltip lists 18 rows, and the axis starts at 0, flattening every move. Each product can be one band from off-peak to peak, with base inside.

Directions considered

- **One chart, with one month focused as a band and the others as grey lines:** always readable, and it fits a question like "how is my Q1 hedge doing?", but only one product shows in detail at a time, and comparing means clicking back and forth.
- **Small multiples, one small chart per product on a shared price scale (chosen):** every product stays readable, with no legend to learn.

Why small multiples

- **Organisation and reading:** each product has its own space, in delivery order, and heights compare directly on the shared scale.
- **Adaptability:** the same design serves month, quarter and year products; granularity changes the number of charts, not the design.
- **Export:** each chart stands alone, so one product or all of them can be exported.
- **Details:** on hover the date shows once, above the grid, and every chart follows the same crosshair. Without both peak and off-peak there is no band, so each profile gets its own line style.
- **Filters:** they keep the live dashboard's controls and live in the URL, so a view can be shared. I considered collapsing the profile chips into "+N" to keep the row on one line, and kept every chip: removing a profile in one click matters more than the space.

Going further

My knowledge of the field is limited, so this rests on an assumption: the page helps customers decide when to hedge, or check existing hedges against the market. Combined with each customer's expected consumption and contract lines, these prices could estimate what the customer will actually pay in the coming months, instead of a price per MWh to translate themselves. It would link Market Data to Contracts and to Budgets.

Responsive and accessibility issues

Noted on the live dashboard, outside the prototype's scope.

- **Primary buttons:** white text on #5D5FEF is 4.83:1, which passes WCAG AA but not AAA (7:1). The hover lightens the background, most likely a 90% opacity, which drops it to about 4.04:1, below AA. Darkening on hover instead fixes it: #383AEB reaches 7.06:1.
- **Contracts toolbar on a narrow window:** the "By contract / By asset" toggle overlaps the filters button, and the search field shrinks until the typed text disappears. Letting the toolbar wrap and giving the search a minimum width would solve both.

Conclusion

The main proposition brings a customer's assets and contracts into one place: a map that locates every site and shows which ones a contract covers, and a site diagram that explains what each asset is and what it is doing.

I am not certain these changes are all needed: as a newcomer, I may not have fully understood how the dashboard behaves or what its customers expect from it. That is why, before going further, I would validate two assumptions with real customers: how common multi-site portfolios are, and what the Futures page is used for.